

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(ii)	$v = \frac{\Delta f c}{2 f_0 \cos 37} \text{ (1) Rearrangement and substitution}$ $v = 0.11[3] \text{ [m s}^{-1}\text{] (1)}$		2		2	2	
	(c)		Any 5 × (1) from: • MRI and ultrasound and CT – good images [only] (1) • X-ray – poor image / soft tissue contrast (1) • Radioactive tracer best or gives more information (1) • Tracer goes to hyperactive [parts of] thyroid (1) • Imaged using gamma camera (1) • Any further detail e.g. - Initial diagnosis using blood (check levels of thyroid hormones) - Iodine gives added contrast (for CT scan) - Tracer can locate problem areas - Tracer can be used to “kill” overactive thyroid / thyroid cancer			5	5		
			Question 8 total	6	9	5	20	10	0